

HRMS v2.0 Migration - TO DO List

Living document - updated as tasks complete. Source of truth for the migration's remaining work.

Milestone status

Completed tasks	26 / 29
Current branch	main @ fc00632
Next task	Phase 5: final cutover to public, then Phase 6 DuckDB decommission
Follow-up progress	6 / 7 done
Phase 0-1 progress	3 / 3 done
Phase 2 progress	2 / 2 done
Phase 3a progress	3 / 3 done
Phase 3b progress	9 / 9 done
Phase 4 progress	3 / 3 done
Phase 5 progress	0 / 1 done
Phase 6 progress	0 / 1 done

TO DO / status by phase

Phase 0-1 (3/3 done)

DONE	Freeze the v1.0 DuckDB schema; inventory tables for ETL Schema freeze note in docs/MIGRATION.md (08f7e40)
DONE	Build v2.0 target schema (db/postgres_schema.sql + Alembic baseline) 50-table public schema, CC-01..CC-16 documented (08f7e40)
DONE	One-time DuckDB -> PostgreSQL ETL with reconciliation hrms DB seeded; counts reconciled (08f7e40)

Phase 2 (2/2 done)

DONE	DuckDB->psycopg adapter (db_backend.py): translate/strftime/autocommit App runs on PostgreSQL legacy schema (b9164f2)
DONE	Unit + browser suites green on PostgreSQL (legacy schema preserved) 29/29 unit, 15/15 Playwright (b9164f2)

Phase 3a (3/3 done)

DONE	Argon2id password hashing; legacy bcrypt re-hashed on login security.py (98563ee)
DONE	Global CSRF enforcement (fetch wrapper + csrf_token field + token API) 98563ee
DONE	Server-side Redis sessions (REDIS_URL opt-in) + login rate limiting 98563ee

Phase 3b (9/9 done)

DONE	CC-01 identity rule enforced: scripts/check_cc_rules.py + PG-gated test 46 identity + 4 natural keys, sequences ahead of data (acb5219)
DONE	Public-flip readiness probe (scripts/probe_public_flip.py) GET + write-flow matrix against a throwaway public DB (acb5219)
DONE	Boolean adapter compat: predicates, INSERT params, naive datetime round-trip Inert on legacy (zero boolean cols); PG-gated tests (21480fe)

DONE	Write-flow probe + salary_structures seed data fix (CC-05) 11/11 core write flows green on public (21480fe)
DONE	CC-09 transactional outbox (outbox.py + scheduler + admin endpoints) Atomic business-write + event; backoff -> dead-letter (7e52f88)
DONE	CC-07 idempotency: @idempotent decorator + idempotency_keys wired for keyed POST retries Replay-without-duplicate, 409 on body reuse, claim released on failure; 6 unit tests green on every stack; probe replays on pure v2.0 JSONB
DONE	Service-layer rewrite inc 1: expanded audit_log (actor/entity/entity_id/before/after/request_id, CC-13) + notifications.category (FR-NOT-03) 8ff66bc; +6 unit tests; DuckDB 46 green
DONE	Service-layer rewrite inc 2: shift_assignments replaces users.shift_start/shift_end (FR-ATT-17); init_db no longer mutates v2.0 public.users get_shift/set_shift helpers reroute ~10 touch points; user CRUD + seed via set_shift; public probe 42/42 write flows
DONE	Extend probe write section: forgot-password, payroll bank-file/TDS, ticket/ATS + verify against a clean public schema Clean hrms_probe re-run: 94/94 GET + 42/42 write flows, including attendance, payroll, and lifecycle paths; all seed-time writes green

Phase 4 (3/3 done)

DONE	Attendance finalisation job (FR-JOB-01) 7 acceptance tests + nightly scheduler + regularization recompute; clean public probe 94/94 GET + 42/42 write
DONE	Maker-checker payroll (FR-PAY-06) Strict state machine + approval trail + Finance/Admin UI; clean public probe 94/94 GET + 42/42 write
DONE	Corrected ATS / onboarding / offboarding flows FR-ATS/FR-ONB/FR-OFF acceptance suite + 94/42 public probe

Phase 5 (0/1 done)

IN PROGRESS	Final cutover: flip APP_DB_SCHEMA to public, retire legacy Disposable rehearsal passed ETL/preflight/94-42 probe/authenticated smoke/CC-01 sequence check; maintenance-window traffic switch pending
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Phase 6 (0/1 done)

PENDING	Decommission DuckDB runtime Out of the ETL blueprint; tracked in SRS
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Follow-up (6/7 done)

DONE	FR-USR archive/restore and session revocation Hard delete replaced with retained archive/restore; admin UI and cross-backend tests updated
DONE	FR-USR directory contract (bounded pagination, sorting, validation) per_page capped at 200, allow-listed sorting, EMP/email/role/departement validation, case-insensitive uniqueness, partial PUT with role-change audit, CSV import under the same contract
DONE	FR-USR-09 permission policy (policy.py matrix + permissions API/UI) 27-module role matrix, deny-beats-allow overrides, full-replace PUT with audit diff, anti-lockout guard, admin permissions modal
DONE	FR-USR-15 policy enforcement wiring (decorators + navbar) Role gates narrowed by policy.can(); navbar derived from the same gate/module tags; department grant modelled; nav/API drift tests
DONE	FR-USR-15 inline scope checks + audited pii_reveal No handler branches on the session role copy; can_view_all is the CC-11 scope half; PII reveal route audited per cross-employee read
DONE	FR-LEA-06/08 policy-derived leave balances Entitlement derived from the effective leave_policy_assignments; reserved ledger enforced; audited policy API + admin modal

PENDING	FR-USR background bulk import jobs, two-person anonymisation Next backend slices after the derived leave balances
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Test & readiness gates

Gate	Stack	Current result
Unit suite (tests/test_app.py)	DuckDB	107 passed, 6 skipped (PG-gated compatibility/public tests)
Unit suite (tests/test_app.py)	PostgreSQL	112 passed, 1 skipped (public-only shift test)
Unit suite (tests/test_app.py)	PostgreSQL + Redis	112 passed, 1 skipped
Browser suite (tests/test_playwright.py)	DuckDB	19 passed (single-threaded server)
Browser suite (tests/test_playwright.py)	PostgreSQL	19 passed
CC-01 rule checker (scripts/check_cc_rules.py)	hrms (public)	OK - 46 identity + 4 natural keys
Public-flip probe (scripts/probe_public_flip.py)	hrms_probe (public)	94/94 GET + 44/44 write flows; lifecycle + permission paths included
Cutover preflight (scripts/cutover_preflight.py)	hrms (public)	Ready: head 0003, identity/sequence rules, required tables, and delta report generated
Disposable Phase 5 rehearsal	hrms_cutover_rehearsal	ETL Phase-1/2 + CC-05 cleanup + preflight + 94/42 probe + authenticated smoke + CC-01 sequence check passed

Update log (newest first)

2026-09-28	FR-LEA-06/08 policy-derived leave balances: the entitlement now comes from the effective leave_policy_assignments (accrual rate x 12, capped by carry-forward) and falls back to the published defaults, so deriving changes nothing until a policy is assigned. The apply path always enforces (it used to skip employees with no row) and reserves the days while a request is pending, which is the first write to the reserved column and closes the double-spending hole. Added an audited policy API plus an admin modal. Also fixed the intermittent DuckDB browser-suite flake (DuckDB attaches a file once per process, so the dev server runs single-threaded there). DuckDB 107 passed / 6 skipped; PostgreSQL 112 passed / 1 skipped; Playwright 19 passed; probe 94/94 GET + 44/44 write.
2026-09-28	FR-USR-15 scope + PII: no handler branches on the session role copy any more (test-enforced), the company-wide list split became policy.can_view_all (CC-11 scope), the dashboard variant follows sees_admin_surface(), and pii_reveal is now enforced and audited: GET /api/users//pii is the only cross-employee personal-data read and every such reveal writes a PII_REVEAL audit row. DuckDB 102 passed / 6 skipped; PostgreSQL 107 passed / 1 skipped; Playwright 18 passed; probe 94/94 GET + 44/44 write.
2026-09-28	FR-USR-15 enforcement wiring: the four role gates now narrow with policy.can() (the role/department check stays the outer gate, so an empty override table is a no-op), every gated view is mapped to a module, and the navbar is derived from the same gate/module tags so a link exists only when the route would admit the user. The HR-department grant is modelled explicitly. Drift removed (Finance no longer sees 403ing links, Super Admin sees the admin links) and /admin/leaves is now reachable. DuckDB 97 passed / 6 skipped; PostgreSQL 102 passed / 1 skipped; Playwright 17 passed; probe 94/94 GET + 44/44 write.
2026-09-28	FR-USR-09 permission policy implemented: policy.py owns the 27-module role matrix with deny-beats-allow per-user overrides, GET/PUT /api/users//permissions does a full replace with an audit diff and an anti-lockout guard, and the admin user list gained a permissions modal. Empty overrides reproduce the role defaults exactly, so no existing authorization outcome changes; decorator/navbar enforcement is the next slice. DuckDB 90 passed / 6 skipped; PostgreSQL 95 passed / 1 skipped; Playwright 17 passed; public probe 94/94 GET + 44/44 write.
2026-09-28	FR-USR directory contract implemented: GET /api/users caps per_page at 200 and sorts through an allow-list (400 on invalid input), POST/PUT validate the employee ID, email, role, department and status with case-insensitive uniqueness, PUT became a true partial update with a before/after audit diff, and the CSV import now validates every row and reports skipped rows. Admin UI mirrors the contract; the public probe stays green.
2026-09-25	Phase 5 cutover rehearsal completed: frozen-source ETL accepted a missing post-v1.0 payroll_approvals table, reconciled Phase-1/Phase-2 data, applied CC-05 cleanup, stamped head, passed read-only preflight and 94/42 public probe, and kept identity sequences ahead after boot seeds and authenticated GET/write smoke. Traffic switch remains a maintenance-window operation.

2026-09-25	FR-ATS/FR-ONB/FR-OFF corrected lifecycle implemented: guarded ATS state machine, 100% offer splits, atomic accepted-offer conversion, signed pre-boarding tokens and real-file validation, five-step guarded onboarding, parallel offboarding with F&F; maker-checker, and IST LWD access revocation, plus lifecycle hardening (strict split precision, ETL preservation, session revocation, encrypted credential delivery, and document authorization). DuckDB 77 passed / 5 skipped; PostgreSQL 81 passed / 1 skipped; Playwright 16 passed; public probe 94/94 GET + 42/42 write.
2026-09-24	FR-PAY-06 maker-checker payroll implemented: Draft → Submitted → Approved → Finalized, Finance/Admin authorization, self-approval rejection, payroll_approvals trail, adjustment-run reference, Finance UI access, and public probe coverage. DuckDB 64 passed / 5 skipped; PostgreSQL 68 passed / 1 skipped; probe 86/86 GET + 30/30 write.
2026-09-24	FR-JOB-01 attendance finalisation implemented: nightly per-shift-date classification, employee weekly-off patterns, holiday/leave precedence, transactional attendance_days replacement, calendar output, and public identity-key probe coverage. DuckDB 64 passed / 5 skipped; PostgreSQL 68 passed / 1 skipped; probe 86/86 GET + 30/30 write.
2026-09-24	Service-layer rewrite inc 2/3 completed: shift_assignments, explicit v2.0 INSERTs, BOOLEAN UPDATE parameter coercion, and binary payroll bank-file export; the clean public-flip probe is fully green.
2026-09-23	CC-07 idempotent writes done: @idempotent decorator on 9 POST routes, idempotency_keys DDL + hourly purge, 6 unit tests green (DuckDB 40 / PG 43 / PG+Redis 43), probe replays on pure v2.0 JSONB -> 86/86 GET + 12/12 write. Next: service-layer rewrite.
2026-09-23	TODO list created. Next task slated: CC-07 idempotency.

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